

Amanda Stoffers

EXTRAGALACTIC ASTROPHYSICS & BAYESIAN INFERENCE METHODS

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Research Profile

I work at the interface of galaxy evolution and statistical methodology. My scientific focus is the evolution of galaxies in the first billion years: what regulated their ionising output, and how did it evolve with star formation and feedback? I study the coupling between star formation, ionising photon production and escape that sets the galaxy contribution to reionisation. Methodologically, I develop fast and flexible inference tools designed to scale to entire *JWST* surveys, including the differentiable, GPU-native spectral energy distribution (SED) fitting software CERIDWEN described below.

Education

- 2023 – Present **PhD in Physics**,
Kavli Institute for Cosmology and Cavendish Laboratory, University of Cambridge, Cambridge, UK
Galaxy Formation and Evolution Group. Supervisor: Dr Sandro Tacchella.
- 2023 – 2024 **Graduate Training Programme, Data Intensive Science**,
University of Cambridge, Cambridge, UK
Advanced coursework in statistics, machine learning and scientific computing.
- 2022 – 2023 **MASt in Physics**,
University of Cambridge, Cambridge, UK
Research project on the escape fraction of ionising photons, First Class. Graduated in the top ten of the cohort.
- 2018 – 2022 **BSc in Physics**, *Technische Universität Berlin*, Berlin, Germany • 1.3
Thesis: *Quantum theory of electron-phonon interaction in gold nanoparticles* (1.0).
German scale, where 1.0 is the highest attainable grade.

Research Software

- 2024 – Present **Ceridwen**, *Author/Lead developer*, github.com/Espe13/ceridwen
A GPU-native, end-to-end differentiable SED-fitting framework in JAX, covering the full forward model from simple stellar populations through nebular emission and dust to projections onto the observed frame. Production inference uses a GPU nested slice sampler, with Hamiltonian Monte Carlo and variational inference available for the same model; supports non-parametric star formation histories of order one hundred age bins with jointly sampled chemical enrichment. Roughly thirty five times faster than CPU PROSPECTOR on high redshift galaxies.
- 2025 – Present **sedpy_jax**, *Author*, github.com/Espe13/sedpy_jax
A JAX reimplement of **sedpy** (B.D. Johnson): GPU-ready, differentiable filter handling and broadband photometric projection – transmission curves, AB and Vega magnitudes, flux-conserving rebinning, air–vacuum conversion, and effective, pivot and zero-point wavelengths across many filters at once. Built to sit underneath CERIDWEN and any gradient-based or simulation-based inference pipeline. MIT licensed.

Publications

First author

- 2026 **The Challenge in Illuminating the Invisible: Constraining LyC Escape with Bayesian Modelling and Symbolic Regression**
Stoffers, A., Tacchella, S., Simmonds, C., Johnson, B. D., & Maiolino, R.
Monthly Notices of the Royal Astronomical Society,
accepted May 2026 (*arXiv:2511.02908*) [[ADS](#)]

First author in prep

expected August 2026 **CERIDWEN: Fast and Flexible GPU-Accelerated Stellar Population Inference**
Stoffers, A., Tacchella, S., Johnson, B. D.

expected September 2026 **JADES: Inferred Ionizing Photon Escape Fractions and the Ionizing Budget of Galaxies**
Stoffers, A., Tacchella, S., Johnson, B. D., Bunker, A., Robertson, B., Simmonds, C., Whitler, S., Witstok, J.

expected September 2026 **JADES: the Role of Star-formation Variability in Shaping the UV Luminosity Function at z 3-9**
Stoffers, A., Tacchella, S., Johnson, B. D., Duan, Q., Robertson, B.

Co-authored

2026 **An (in)complete NIRSpect census of Balmer absorption in Type 1 AGN: radiation-driven outflows in little red dots, quasars and variable stars**
Juodžbalis, I., Ji, X., D'Eugenio, F., Scholtz, J., Maiolino, R., Stoffers, A., et al. (20 authors; 6th)
Submitted (arXiv:2607.19057) [ADS]

2025 **Bursting at the seams: the star-forming main sequence and its scatter at $z = 3-9$ using NIRCам photometry from JADES**
Simmonds, C., Tacchella, S., McClymont, W., Curtis-Lake, E., D'Eugenio, F., et al., incl. Stoffers, A. (27 authors; 11th)
Submitted (arXiv:2508.04410) [ADS]

2024 **Ionizing properties of galaxies in JADES for a stellar mass complete sample: resolving the cosmic ionizing photon budget crisis at the Epoch of Reionization**
Simmonds, C., Tacchella, S., Hainline, K., Johnson, B. D., Puskás, D., et al., incl. Stoffers, A. (29 authors; 23rd)
Monthly Notices of the Royal Astronomical Society, 535(4), 2998–3019 [ADS]

Invited Talks

2026 **Invited seminar**, Friday Late Afternoon Talks, UCL, London, UK
Differentiable Galaxies: What Gradient-Based Inference Changes About SED Fitting

2025 **Invited talk**, JADES Science Focus Meeting, Santa Cruz, CA, USA
CERIDWEN: Fitting JADES Star Formation Histories at Survey Scale on GPUs

Contributed Talks and Presentations

2026 **Contributed talk**, Simulation Based Inference for Galaxy Evolution, Cambridge, UK
An Explicit-Likelihood Alternative to SBI: Differentiable Stellar Population Synthesis in JAX

2025 **Contributed talk**, LyC Escape from Galactic Labyrinths, Crete, Greece
Inferring Lyman Continuum Escape Fractions via SED Modelling

2025 **Contributed talk**, Kaba Kada: Exploring the First Billion Years of the Universe, Port Douglas, Australia
The Challenge in Illuminating the Invisible: Constraining LyC Escape with Bayesian Modelling and Symbolic Regression

2025 **Workshop presentation**, IAIFI Summer School and Workshop, Harvard University, Boston, MA, USA
Contextual Bandits for Stroke Treatment Analysis

- 2024 **Workshop presentation**, *IAIFI Summer School and Workshop*, MIT, Boston, MA, USA
Anomaly Detection via Latent Space Evaluations
- 2024 **Contributed talk**, *National Astronomy Meeting (NAM)*, Hull, UK
Lyman Continuum Escape Fractions of High-Redshift Galaxies

Grants, Awards and Scholarships

- 2023 – 2027 **STFC Data Intensive Science / Cavendish Scholarship**, Cambridge, UK
 Full doctoral funding.
- 2025 **ESA-funded research visit**, *JADES Science Focus Meeting*, Santa Cruz, CA, USA
- 2025 **Postgraduate Research Fund**, *Jesus College, Cambridge*
 Travel award to present at *Kaba Kada: Exploring the First Billion Years of the Universe*, Port Douglas, Australia.
- 2025 **Research Culture Grant**, *Kavli Institute for Cosmology*, Cambridge, UK
 Awarded to establish and run the Astro Career Talk Series.
- 2025 **Best Talk and Presentation**, *Graduate Research Conference*, *Jesus College*, Cambridge, UK
- 2021 – 2023 **Studienstiftung des deutschen Volkes**, *German Academic Scholarship Foundation*
 Germany's national academic scholarship, awarded to roughly the top one per cent of students.
- 2021 **Enable AI Scholarship**, *Data science and machine learning training*
- Earlier KMK Scholarship, exchange programme, Trentino, Italy (2017); Mathematics Abitur Award, Deutsche Mathematiker-Vereinigung (2018); 1st place, Berlin regional Mathematics Olympiad (2014).

Teaching and Supervision

- 2023 – 2025 **Supervisor**, *Relativistic Astrophysics and Cosmology*, Cavendish Laboratory, University of Cambridge
 Small-group undergraduate supervisions, run in each of the 2023, 2024 and 2025 academic years.
- 2023 **Supervisor**, *Mathematics for Natural Sciences, Part IA*, Trinity Hall, University of Cambridge
 First-year small-group supervisions.

Professional Service and Leadership

- 2026 **Local Organising Committee**, *Hubble and Beyond: Morphology and Kinematics of Early Galaxies*, Kavli Institute for Cosmology, Cambridge
- 2024 – 2025 **Founder and Committee Member**, *Astro Career Talk Series*, Kavli Institute for Cosmology, Cambridge
 Established and ran a departmental series of interviews with academic and industry researchers; secured a Research Culture Grant to fund it.
- 2024 – 2025 **Committee Member**, *Cambridge University Women in Physics Society*
- 2024 **Local Organising Committee**, *Star Formation 2024*, Kavli Institute for Cosmology, Cambridge
- 2022 – 2023 **Student Representative**, *MASt Physics Programme*, University of Cambridge
- 2021 – 2025 **Board Member**, *In.Ph.A.S.E. Student Association*
 Organised the student colloquium programme.
- 2024 – Present **Jesus College Boat Club**, University of Cambridge
 W1 Captain (2024–2025), leading and coaching the college's senior women's crew; Committee Member (2026–present).

Outreach and Public Engagement

- 2018 – 2023 **Presenter and Event Host**, *Planetarium Berlin*, Berlin, Germany
Public astronomy talks and show presentation for general audiences.
- 2019 – 2022 **Volunteer Instructor**, *Observatories e.V.*, Berlin, Germany
Ran observing sessions and hands-on experiments for public observatory programmes.
- 2021 – 2022 **Student Laboratory Tutor**, *DESY Zeuthen*, Zeuthen, Germany
Supervised school students running particle-physics experiments.

Other Professional Experience

- 2023 **Quantum Software Engineer**, *Diehl Defence*, Überlingen, Germany
Implemented quantum algorithms and developed the associated mathematical models.
- 2021 – 2022 **Educational Assistant**, *Grunewald Primary School*, Berlin, Germany

Technical Skills

- Inference Nested sampling, Hamiltonian Monte Carlo, variational inference, hierarchical Bayesian modelling, symbolic regression, automatic differentiation
- Computing Python (JAX, NumPyro, PROSPECTOR, Qiskit, QuTiP), C/C++, GPU programming, Git, Linux, high-performance computing, L^AT_EX
- Astronomy *JWST* NIRCam and NIRSpect data, stellar population synthesis, joint photometric–spectroscopic SED modelling
- Collaborations JADES (*JWST* Advanced Deep Extragalactic Survey); LAPIS
- Languages German (native), English (near-native), French (B2), Italian (A2), Spanish (A1)